Global Patterns in Climate Awareness, COVID-19 Impact, and National Happiness

Exploring Disparate Data: Part 3 - Final Report

The Geese

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1 Data Description

1.1 Climate Change Awareness Data

```
## # A tibble: 110 x 7
##    country         aware_no aware_alittle aware_moderate aware_alot aware_refuse
##    <chr>              <dbl>         <dbl>          <dbl>      <dbl>        <dbl>
##  1 Albania             9.99          43.0           33.1      10.6         3.25
##  2 Algeria            23.2           37.6           26.0      11.1         2.15
##  3 Angola             19.1           43.0           23.5      11.1         3.29
##  4 Argentina           8.25          37.4           43.8       9.88        0.696
##  5 Armenia             9.18          51.4           25.8      11.4         2.14
##  6 Asian & Pacifi~    13.3           40.1           24.8      19.5         2.26
##  7 Australia           1.13          26.4           48.1      23.9         0.407
##  8 Austria             1.34          16.4           48.6      32.7         0.985
##  9 Azerbaijan          6.19          42.4           35.9      13.2         2.20
## 10 Bangladesh         24.5           36.2           16.2      18.9         4.22
## # i 100 more rows
## # i 1 more variable: aware_base <dbl>
```

The data come from HDX (Humanitarian Data Exchange) and describes the responses to a survey in 2022 from the Meta Climate Change Opinion Survey. This survey asked people in many different countries their level of awareness on climate change and recorded their responses [1]. The people doing the survey had the option to choose between different statements that represented their awareness of climate change. For example, if they we not aware at all the participant would select "I have never heard of it".

In order to clean the data, we first adjusted so that all of the periods in between name were removed and replaced with spaces. Then, we adjusted the country names that were called something different from their name in the other data sets to match the other data sets country names. This was to ensure that in joining everything goes smoothly.

[1] Meta. (2022). Meta Climate Change Opinion Survey 2022. Humanitarian Data Exchange (HDX). https://data.humdata.org/dataset/meta-climate-change-opinion-survey

1.2 Covid 2020 Cases Data

```
## # A tibble: 104 x 9
##    country    aware_no aware_alittle aware_moderate aware_alot aware_refuse
##    <chr>         <dbl>         <dbl>          <dbl>      <dbl>        <dbl>
##  1 Albania        9.99          43.0           33.1      10.6         3.25
##  2 Algeria       23.2           37.6           26.0      11.1         2.15
##  3 Angola        19.1           43.0           23.5      11.1         3.29
##  4 Argentina      8.25          37.4           43.8       9.88        0.696
##  5 Armenia        9.18          51.4           25.8      11.4         2.14
##  6 Australia      1.13          26.4           48.1      23.9         0.407
##  7 Austria        1.34          16.4           48.6      32.7         0.985
##  8 Azerbaijan     6.19          42.4           35.9      13.2         2.20
##  9 Bangladesh    24.5           36.2           16.2      18.9         4.22
## 10 Belgium        1.40          26.9           57.0      14.6         0.0961
## # i 94 more rows
## # i 3 more variables: aware_base <dbl>, continent <chr>, total_cases <dbl>
```

The data come from the Our World in Data COVID_19 database and detail the number of covid cases in each country in 2020, recorded daily [2].

In order to clean the data, we adjusted some of the names to match the climate awareness data by changing some of the names of certain countries so that they all match.

1.3 World Happiness Report Data

```
## # A tibble: 97 x 10
##    country    aware_no aware_alittle aware_moderate aware_alot aware_refuse
##    <chr>         <dbl>         <dbl>          <dbl>      <dbl>        <dbl>
##  1 Albania        9.99          43.0           33.1      10.6         3.25
##  2 Algeria       23.2           37.6           26.0      11.1         2.15
##  3 Angola        19.1           43.0           23.5      11.1         3.29
##  4 Argentina      8.25          37.4           43.8       9.88        0.696
##  5 Armenia        9.18          51.4           25.8      11.4         2.14
##  6 Australia      1.13          26.4           48.1      23.9         0.407
##  7 Austria        1.34          16.4           48.6      32.7         0.985
##  8 Azerbaijan     6.19          42.4           35.9      13.2         2.20
##  9 Bangladesh    24.5           36.2           16.2      18.9         4.22
## 10 Belgium        1.40          26.9           57.0      14.6         0.0961
## # i 87 more rows
## # i 4 more variables: aware_base <dbl>, continent <chr>, total_cases <dbl>,
## #   ladder_score <dbl>
```

This data comes from the World Happiness Report from 2023 and detail how people responded to a question about the “Life Ladder” [3] about how they rate their overall life satisfaction from 0 to 10. The data has the overall well being of a country depending on how participants from these countries rated their life on the ladder.

We didn’t have to do any cleaning for this data, as it was already clean from part 1!

[2] Ritchie, H., Mathieu, E., Rodés-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J., Macdonald, B., Beltekian, D., & Roser, M. (2020). Coronavirus pandemic (COVID-19). Our World in Data. https://ourworldindata.org/coronavirus

[3] Helliwell, J. F., Layard, R., Sachs, J. D., & De Neve, J.-E. (2023). World Happiness Report 2023. Sustainable Development Solutions Network. https://worldhappiness.report

1.4 Combining the Data

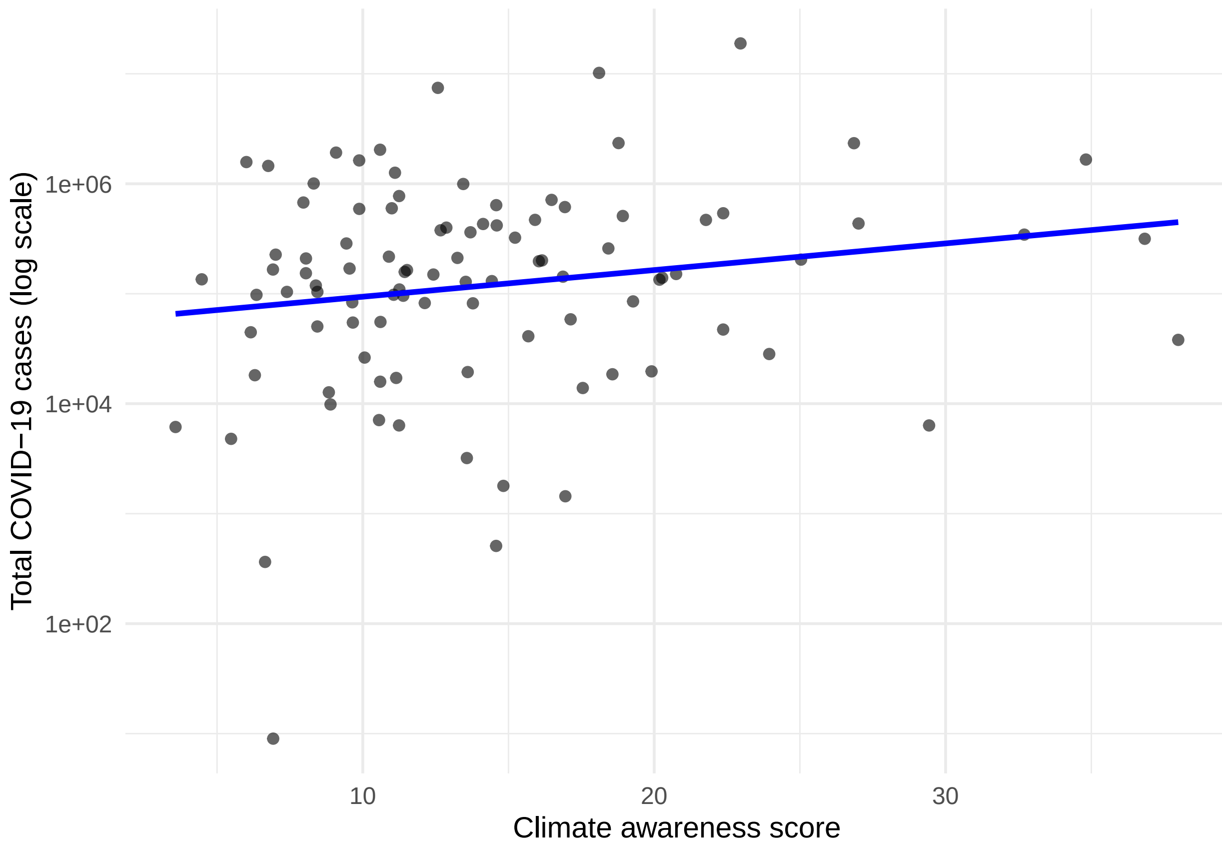
We first joined climate awareness data to covid data by selecting only columns "country", "continent", and `total_cases`. Then, we used `inner_join` to join those columns of the covid 2020 data with the climate awareness data.

For combining happiness data with climate awareness and covid data, we joined this data to the data set `climate_covid`, which was made from joining the previous two data sets. We used `inner_join()` to join the columns `"country"` and `"ladder_score"` from the happiness data to the climate awareness and covid data.

2 Exploratory Data Analysis

We explored many aspects of the data, but will demonstrate three. These are countries with higher covid cases there are more people reporting high climate awareness, countries with higher climate awareness are happier, and countries with higher covid cases are also happier

The first aspect that we found interesting is shown in this scatterplot. Figure 1.



We see a positive relationship between the percentage of respondents who reported "I know a lot about climate change" and the total number of COVID-19 cases [4][5]. This pattern does not imply that climate awareness causes higher case count. Rather, it reflects broader national characteristics. Moreover, countries with stronger education systems and greater access to information tend to show higher levels of climate

[4] Meta. (2022). Meta Climate Change Opinion Survey 2022. Humanitarian Data Exchange (HDX). https://data.humdata.org/dataset/meta-climate-change-opinion-survey

[5] Ritchie, H., Mathieu, E., Rodés-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J., Macdonald, B., Beltekian, D., & Roser, M. (2020). Coronavirus pandemic (COVID-19). Our World in Data. https://ourworldindata.org/coronavirus

awareness. These same countries also typically have higher population densities, more international travel, and more comprehensive COVID-19 testing and reporting infrastructure. Together, these factors produce the positive trend observed in the data.

This insight is supported by the summary statistics in table 1.

metric	correlation
Correlation	0.175

The next insight that we found is shown in figure 2.

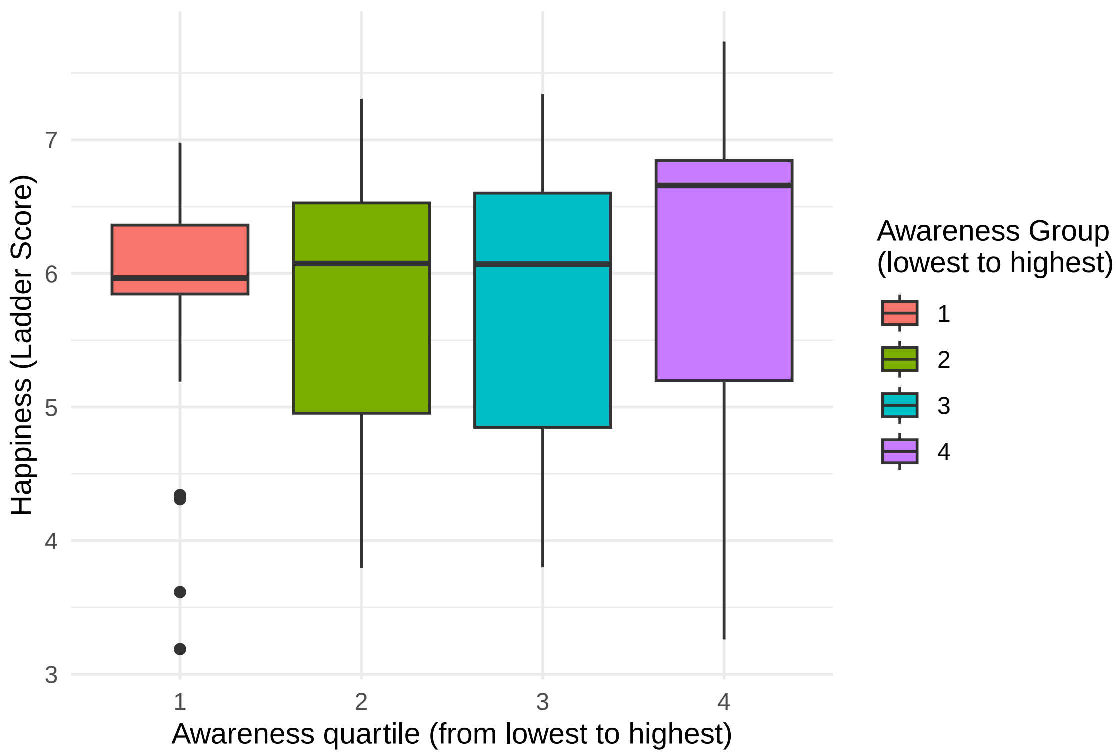


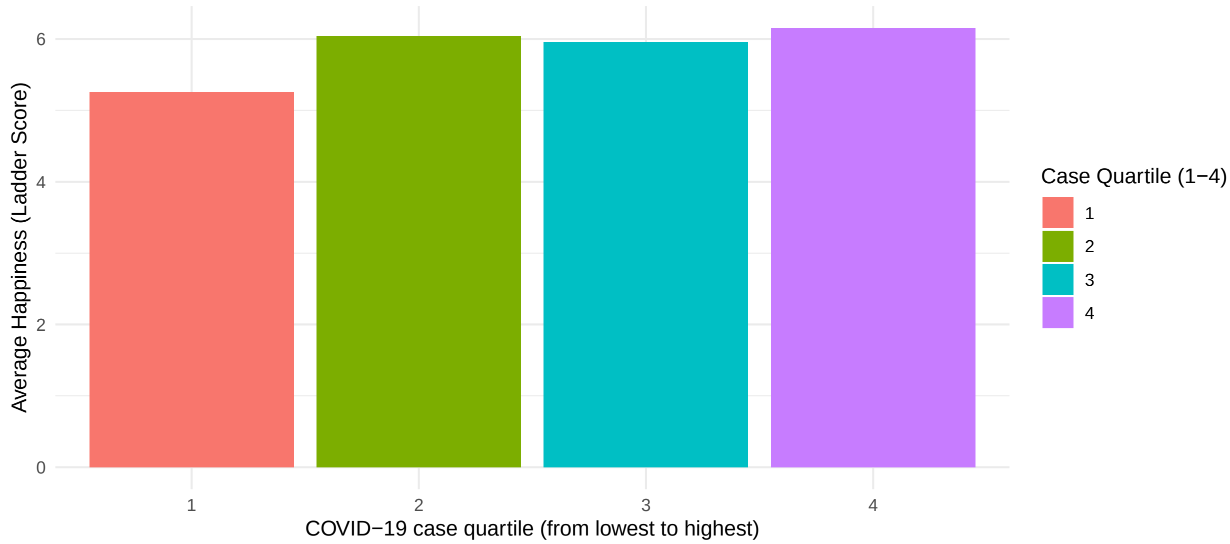
Figure 1: Happiness scores by climate awareness quartile

Countries with greater climate awareness tend to have higher reported happiness scores, based on data from the Meta Climate Change Survey and the World Happiness Report [6 7].

This boxplot shows a clear upward trend, indicating that countries with greater climate awareness tend to have higher reported happiness scores. This pattern likely reflects broader structural differences rather than a direct causal relationship. The country with higher climate awareness often also have better access to education, higher GDP, more stable institutions, and stronger scientific and media communication systems. These underlying factors contribute both to increased awareness of climate issues and to higher Life Ladder scores, which helps explain the positive association observed in the data.

[6]Meta. (2022). Meta Climate Change Opinion Survey 2022. Humanitarian Data Exchange (HDX). https://data.humdata.org/dataset/meta-climate-change-opinion-survey

[7]Ritchie, H., Mathieu, E., Rodés-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J., Macdonald, B., Beltekian, D., & Roser, M. (2020). Coronavirus pandemic (COVID-19). Our World in Data. https://ourworldindata.org/coronavirus



Countries in the highest COVID-case quartile also report the highest average happiness, which reflects patterns seen in COVID-19 reporting and Life Ladder data [8,9].

Surprisingly, these countries in the highest COVID-case quartile also report the highest average happiness. This counterintuitive pattern can be explained by the fact that regions such as Western Europe, North America, Australia, and parts of East Asia both experienced high reported case counts and consistently rank among the highest on the Life Ladder. In contrast, many countries with limited testing capacity reported fewer cases but also tend to have lower happiness scores. As a result, the association between higher case counts and higher happiness is indirect and largely reflects differences in economic development and infrastructure rather than any meaningful effect of COVID-19 itself.

3 Conclusion and Future Work

In summary, we identified that there are three patterns that emerge from the different datasets. Firstly, a correlation exists between greater numbers of reported COVID-19 infections and greater levels of climate awareness. This is presumably a factor of better education, scientific communication, and reporting ability that is found in developed areas. Secondly, a positive correlation exists between climate awareness and national levels of happiness. This finding is indicative of climate awareness being a symptom of a wider set of advantages that can be directly linked to greater well-being. Finally, a positive correlation exists between the provinces that experience the highest numbers of COVID-19 infections and the happiest provinces. This finding is most likely a symptom of differing levels of development, as provinces that test regularly report a higher Life Ladder. In aggregate, these findings reinforce that a range of different factors, such as health, climate, and well-being, interact through underlying structures of a country's socioeconomic system [10,11].

For future research, a range of additions could be made to improve this analysis. Including the likes of GDP, literacy, or something of that sort to index for development can index for development, & this can be analyzed to better understand the relationships found above [12,13]. Multiple regression can be done to

[8] Ritchie, H., Mathieu, E., Rodés-Guirao, L., Appel, C., Giattino, C., Ortiz-Ospina, E., Hasell, J., Macdonald, B., Beltekian, D., & Roser, M. (2020). Coronavirus pandemic (COVID-19). Our World in Data. https://ourworldindata.org/coronavirus

[9] Helliwell, J. F., Layard, R., Sachs, J. D., & De Neve, J.-E. (2023). World Happiness Report 2023. Sustainable Development Solutions Network. https://worldhappiness.report

[10] The World Bank. (2023). Country and lending groups. https://datahelpdesk.worldbank.org/knowledgebase/articles/906519

[11] United Nations Development Programme. (2023). Human Development Report 2023. https://hdr.undp.org

[12] The World Bank. (2023). Country and lending groups. https://datahelpdesk.worldbank.org/knowledgebase/articles/906519

[13] United Nations Development Programme. (2023). Human Development Report 2023. https://hdr.undp.org

test for direct & indirect relationships among the variables. An exploration into whether high awareness of climate predicts high desire to combat climate, or through policies, can also be done. Adding population, or something of that sort, such as COVID-19 cases relative to population/cap, can make comparisons better among differing-sized countries.

However, this analysis contains a number of caveats. Climate awareness metrics are subject to survey answer biases, which could be contingent on sampling quality and surveyee integrity. Discrepancies within the data, such as country name errors, require manual preprocessing, potentially leading to a country being left out. The reporting of COVID-19 case information is subject to testing capacity variability, such that countries that have greater capabilities tend to report a larger number of infections simply because of a larger testing scope. Notably, these patterns are better understood as correlational than causational, where a underlying structure is suggested that is independent of variable interaction. Finally, the sizes of the sampling sets tend to differ by country. Despite these constraints, our results have identified important and interpretable trends for climate literacy, health, and well-being across the world. By combining multiple sets of data, this study offers a unique lens through which the social, economic, and institutional forces that drive trends of awareness, happiness, and health data can be viewed.